Assessing the Credibility of Organisational Knowledge from Fragmented Data

1 Introduction

Company information is spread across policies, emails, meetings and chat messages. Sources can be outdated, repeated or contradictory.

Research Question

How can we assess a claim when company records are incomplete or disagree?

Project Aim

Give each claim a clear verdict, a credibility score and traceable evidence.

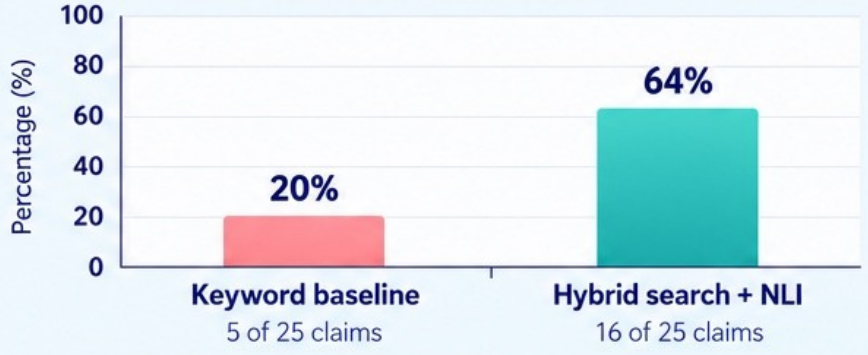
2 Methods

1. **Find evidence**
   Search company records for relevant passages.
2. **Check agreement**
   Use Natural Language Inference (NLI) to identify support, contradiction or unrelated text.
3. **Weigh and score**
   Consider authority, recency and independence. Combine the evidence into a credibility score.
4. **Explain the verdict**
   Show the result, uncertainty and cited evidence.

3 Results

Evaluation Results on Gold Claims

Compared with human-labelled answers for 25 claims.



20%
Keyword Baseline
5 of 25 exact verdicts

84%
Directional Agreement
21 of 25 claims

3.2×
Exact-Match Improvement
64% ÷ 20%

Exact agreement: the predicted verdict matches the human label.

Directional agreement: positive verdicts are grouped together, and negative verdicts are grouped together; conflicted and insufficient-evidence verdicts remain separate.

Better evidence interpretation improved verdict accuracy from 20% to 64%.

Initial evaluation on 25 claims; broader testing is needed.

4 Worked Example

Claim

"Discounts above 15% require Finance approval before a deal is closed."

10 supporting passages

2 contradicting passages

SUPPORTED
Credibility score: **0.994**
High confidence · 90% interval: 0.95–1.00

Supporting evidence outweighs conflicting records, including sources that refer to an older rule.

5 Conclusions

- Shows why a claim should be trusted.
- Makes contradictions and evidence gaps visible.
- Helps people make defensible, evidence-based decisions.

Source: Project presentation, slides 10–15 and 17.